



**Model Question Paper Computer Science HSSC
(COMPOSITE)
FOR ANNUAL EXAMINATION 2026 & ONWARDS
Practical Based Assessment (PBA)
Computer Science Curriculum (2022-23)-Scheme of Studies 2009**



Total Marks: 50

Time: 2 hours 30 minutes

Note: Attempt all questions and write answers within provided spaces on the E-sheet.

SECTION-A (30 Marks)

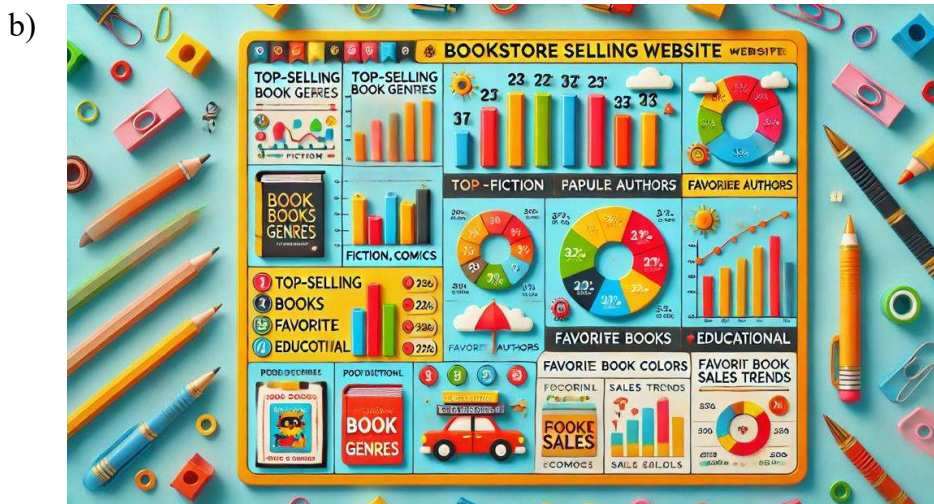
Q No.1.

[10 marks]

- a) Write a Python code that uses a function to enter a number, square it, and continue reading numbers until a negative number is entered. [5 marks]
- b) You are an entrepreneur who wants to start an online T-shirts store. Your goal is to create an MVP (minimum viable product) to quickly test your idea with potential customers.
- i. Identify the most essential feature for the online store. [1 mark]
 - ii. Identify an appropriate tool and technology you will use to create the Frontend and Backend. [1 mark]
 - iii. Once the MVP is complete, gather feedback from potential users (friends, family, etc.) and plan future improvements. Design any three future improvements. [3 marks]

OR

- a) Write a Python program that opens the file and reads its content. Prints the content of the file line by line. [5 marks]



Here is a prototype for an online bookstore. Provide feedback on how to enhance its design and functionality. [5 marks]

- i) What do you think is the strongest part of the design?

- ii) What changes would you recommend making the prototype more engaging?
- iii) Are the labels easy to read and understand? Justify your answer.
- iv) Would this chart help students learn about organizing and interpreting data?
- v) Is there any data you think should be added to make the chart more relatable for students?

Q No.2. You are a weather scientist tracking the rainfall (in mm) for each month of the year. Use Python list to analyze and display this data. **[10 marks]**

- a) Store the rainfall data for each month in a Python list and print it. [2 marks]

Use the following data (in mm): [50, 80, 60, 30, 120, 90, 70, 100, 85, 95, 40, 65]

- b) What does the following Python code prints? [1+1 marks]

```
rainfall=[50, 80, 60, 30, 120, 90, 70, 100, 85, 95, 40, 65]
print(max(rainfall))
print(min(rainfall))
```

- c) Examine the following Python program, identify any errors, and correct them. [4 marks]

```
rainfall=[50, 80, 60, 30, 120, 90, 70, 100, 85, 95, 40, 65]
avg= sum(rainfall)/12;
print("average rainfall = , average)
```

- d) Modify the following Python code such that it prints the months with "**Dry**" rainfall. Also print the months with dry rainfall. [2 marks]

```
rainfall=[50, 80, 60, 30, 120, 90, 70, 100, 85, 95, 40, 65]
count=0
for i in range(10):
    if rainfall[i] >=70:
        count= count + 1
print("total months with greater than 70mm rainfall: ", count)
```

Q No.3. A college teacher wants to track the attendance of students in a class: **[10 marks]**

- i. Write a Python program that asks the user whether a student is present or absent. [2 marks]

If the student is "present," print "Student is present."

If the student is "absent," print "Student is absent."

- ii. Modify the above program to enter attendance for **five (05)** students. [3 marks]

- iii. After tracking the attendance of all five students, print the total number of students **present**. [3 marks]

- iv. Add a feature to calculate and print the percentage of students who are **present**. [2 marks]

SECTION-B (20 Marks)

Q No.4.

[6 marks]

- a) Write an algorithm to input three numbers and prints the smallest number. [3 marks]
- b) Write a Binary search algorithm to find the number **36** in the given list: [3 marks]

93	87	83	71	65	52	47	36	10
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OR

- a) Develop a pseudocode that prints even numbers from 1 to 100. [3 marks]
- b) Draw a trace table for the given pseudocode: [3 marks]

Step 1: START
Step 2: num1 = 3, num2 = 4
Step 3: total ← 0
Step 4: FOR i FROM 1 TO num2 DO
Step 5: total ← total + num1
Step 6: END FOR
Step 7: PRINT total
Step 8: END

Q No.5.

Simplify the Boolean Function F using the Karnaugh Map and also construct the logic circuit for the simplified expression.

[4+2 marks]

$$F = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC + \bar{A}BC$$

Q No.6.

[8 marks]

- a) Write a python code to generate a dataset with two variables where $y=3x+4$ and plot a line chart and a box plot. [2+2 marks]
- b) i) You are assigned to explore the reasons behind students' preferences for online learning compared to in-person learning. What kind of questions would you ask in your interview to gather qualitative data on students' learning preferences (any four questions)? [2 marks]
- ii) You are conducting a survey to understand students' reading habits. Distribute the survey to 50 students in your class. Design a questionnaire having at least four appropriate questions to collect data about students' reading habits with closed-ended questions. [2 marks]